# Project Design Phase-II
# Technology Stack (Architecture & Stack)

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| Date | 31 January 2026 |
| Team ID | Vidhya Jain |
| Project Name | Global Fuel Prices Analysis |
| Maximum Marks | 2 Marks |

## Define Your Technology Stack:

Identify the technologies, platforms, databases, BI tools, and integrations that you will use to build your Business Intelligence solution. A well-chosen technology stack helps ensure that the solution is scalable, maintainable, secure, and suitable for data analysis and reporting requirements.

| S.No | Component | Description | Technology | Justification / Purpose |
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| 1. | Programming / Query Language | Language used to query, filter, transform, aggregate, and analyze data. | Python, SQL, HTML, CSS | SQL queries the fuel price data in Snowflake; Python (pandas) is used for cleaning and checks; HTML/CSS for the web front-end layout. |
| 2. | Database / Data Warehouse | Warehouse Used to store and manage structured business data for analysis and reporting. | Snowflake / Excel / CSV | Fuel price data (country, fuel type, date, price in USD/litre) is stored in structured tables and files, giving one reliable source for Tableau. |
| 3. | Data Ingestion | Used to load data from source files or external systems into the analytical environment. | Snowflake Stages / COPY INTO / Tableau Data Connections | CSV files are loaded into Snowflake with Stages and COPY INTO; Tableau connects directly to the data for fast, repeatable refresh. |
| 4. | Data Transformation and Business Logic | Tools/Logic used to clean, join, filter, aggregate, data for analysis and calculate KPIs, metrics, business rules, and analytical measures. | Tableau Prep, Snowflake SQL, Tableau Desktop | Clean missing values and duplicates, standardise country names, and build calculated fields for average price, YoY change, ranking and petrol-diesel gap. |
| 5. | Data Visualization / Analytics | Used to create interactive dashboards, charts, reports, KPIs, and visual analysis. | Tableau, Python | Tableau gives the world map, trend lines, top/bottom country charts and KPI cards with filters; Python supports extra checks and statistics. |
| 6. | Front-End / User Interface | Interface through which users interact with dashboards, filters, parameters, and visual insights. | Tableau Public | Tableau Public lets users open the dashboard in a browser, use filters, parameters and tooltips, and share it with a link without installing software. |